Challenges Overview

tracks

Future Mobility (Automotive)

Industry & Manufacturing
(Operations & Logistics)

Advanced Robotics
(Automation & Maintenance)

Cognitive Platforms
(Health, Defense, Home & Support)

challenges

Future Mobility (Automotive)

GlobalGate AI: Solution that audits Advanced driver-assistance system

Agentic Basic Software Modules Coding: Agents to configure Automotive Open System Architecture (AUTOSAR)

Intelligent Observability & Self-Healing: Engineer a self-healing telemetry platform that ingest multi-environment data streams.

Voice Assistant for Vehicle Control: That intelligently splits workloads between edge and cloud to control for reduced latency

Industry & Manufacturing

Concept Critic: Agentic solution designed to systematically audit supplier blueprints, identifying technical gaps and structural over-engineering

MachineWhisperer: AI assistant that instantly serves the exact troubleshooting solution directly to the shop floor

Advanced Robotics

FarmTwin: Advanced farm machinery operates in a complete data silo, isolated from the live conditions of the field it is working on. Synchronize real-time tractor and harvester telemetry with live field conditions to enable automated, condition-triggered work orders.

Cognitive Platforms

The Sentient Home: Use LLMs and Home Connect APIs to build a proactive companion that anticipates routines and adapts to your daily mood.

SynergyEngine: Massive corporate mergers leave invaluable business data trapped across fragmented, silos. Build a smart data harmonization engine to unify post-merger datasets.

Challenge - GlobalGate AI

Future Mobility
(Automotive)

GlobalGateAI

Develop a domain-agnostic SaaS platform that automates cross-regional compliance for ADAS software (Advanced Driver-Assistance System software), eliminating repetitive development cycles.

Problem Statement: Moving automotive software to a new global region usually means starting compliance checks entirely from scratch.

Mission: Build a unified solution that instantly audits ADAS software for regional compliance to slash deployment times.

Business Impact: Cost reduction, Efficiency improvements, Improved customer or user experience. Domain agnostic AI compliance check and can be provided service for revenue generation.

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Challenge - Agentic BSW Coding

Future Mobility
(Automotive)

Agentic Basic Software Modules Coding

Agentic BSW (Basic Software Modules) Coding: Leverage agentic AI architectures to autonomously configure AUTOSAR-conforming basic software directly from complex system requirements.

Problem Statement: Configuring AUTOSAR (Automotive Open System Architecture) basic vehicle software is highly complex, tedious, and manual.

Mission: Build autonomous AI agents that read system requirements and automatically configure the Cubas framework.

Business Impact: Cost reduction, Efficiency improvements, Improved customer or user experience, Better operational visibility or insights

Challenge - Intelligent Observability & Self-Healing

Future Mobility
(Automotive)

Intelligent Observability & Self-Healing

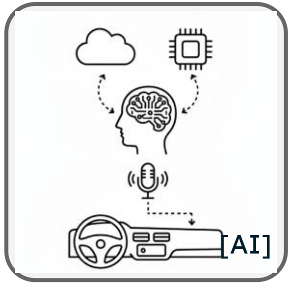
Engineer a self-healing telemetry platform that correlates multi-environment cloud and ECU logs to predict and mitigate critical downtime autonomously.

Problem Statement: When automotive cloud systems experience failures, technicians drown in a sea of siloed logs while downtime costs tick away.

Mission: Ingest massive multi-environment data streams to detect anomalies in real-time and trigger automated self-healing actions.

Business Impact: Cost reduction, Efficiency improvements, Better operational visibility or insights

Challenge - Voice Assistant for Vehicle Control



Future Mobility
(Automotive)

Voice Assistant for Vehicle Control

Build a hybrid AI-powered in-vehicle voice assistant that intelligently routes commands between edge and cloud systems to enable fast, reliable, real-time vehicle control.

Problem Statement: In-vehicle voice control today relies entirely on cloud connectivity / introducing latency and single points of failure for safety-critical commands.

Mission: Build a hybrid voice assistant that intelligently splits workloads between edge and cloud to control vehicle APIs in real-time / using agentic AI patterns for latency-aware decision-making.

Business Impact: Improved customer or user experience, Efficiency improvements

Challenge - Concept Critic

Industry & Manufacturing
(Operations & Logistics)

Concept Critic

Construct a agentic solution designed to systematically audit supplier blueprints, identifying technical gaps and structural over-engineering.

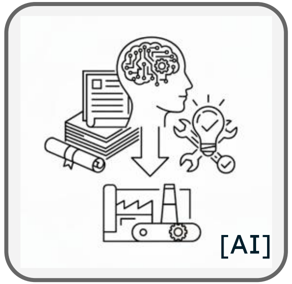
Problem Statement: Vetting supplier blueprints for factory machinery takes days of expert effort to spot hidden design flaws or over-engineering. The harder question is rarely asked systematically: is this the right engineering answer to the actual task?

Mission: Build a multi-agent AI system that acts as a “Devil’s Advocate” to systematically interrogate and critique machinery proposals, flagging both over-engineering and gaps, machinery-agnostic and traceable.

Business Impact: Cost reduction, Efficiency improvements

Challenge - MachineWhisperer

Industry & Manufacturing
(Operations & Logistics)



MachineWhisperer

Optimize manufacturing maintenance efficiency by building an AI-assistant (and/or RAG-powered interface) that synthesizes dense technical documentation into instant, actionable troubleshooting guidance.

Problem Statement: Every minute an assembly line stands still costs thousands, yet technicians waste vital time hunting for fixes in 400-page manuals.

Mission: Create an interactive AI assistant that instantly serves the exact troubleshooting solution directly to the shop floor.

Business Impact: Cost reduction, Efficiency improvements, Improved customer or user experience, New digital products or services, Better operational visibility or insights

Challenge - FarmTwin

Advanced Robotics
(Automation & Maintenance)

FarmTwin

Synchronize multi-source vehicle telemetry with localized environmental data models to generate condition-triggered, autonomous work orders. A real-time Digital Twin ecosystem for farms, connecting soil, crops, water, machinery, and weather into one unified operational model.

Problem Statement: Advanced farm machinery operates in a complete data silo, isolated from the live conditions of the field it is working on. The gap is integration. Each layer generates data. None of them talk to each other in a structured, semantically meaningful way that allows a farm operator to ask the fundamental question:

Mission: Synchronize real-time tractor and harvester telemetry with live field conditions to enable automated, condition-triggered work orders. *"Given everything happening on my farm right now, soil, crops, weather, and machines, what should I do next, and what happens if I don't?"*

Business Impact: Yield optimization, Water savings, Machinery efficiency, Risk reduction, Cost reduction, New revenue, Sustainability contribution.

Challenge - The Sentient Home

Cognitive Platforms
(Health, Defense, Home & Support)

The Sentient Home

Advance smart home architecture by utilizing Large Language Models to transform connected hardware into a context-aware, proactive living environment.

Problem Statement: Smart homes today are just a collection of connected, manual buttons rather than adaptive environments.

Mission: Use LLMs and Home Connect APIs to build a proactive companion that anticipates routines and adapts to your daily mood. transform the smart home from a collection of connected devices into a truly "sentient" entity. The goal is to create an AI-powered home companion that understands, anticipates, and adapts to the user's needs and lifestyle. This goes beyond simple voice commands, aiming for a proactive, personalized, and deeply integrated home experience.

Business Impact: Improved customer or user experience, New digital products or services

Challenge - SynergyHarvester

Cognitive Platforms
(Health, Defense, Home & Support)

SynergyHarvester

Build a smart data harmonization platform to unify post-merger datasets across product portfolios and instantly unlock hidden cross-selling opportunities.

Problem Statement: Massive corporate mergers leave invaluable business data trapped across highly fragmented, incompatible software ecosystems.

Mission: Build a smart data harmonization engine to unify post-merger datasets and immediately unlock hidden cross-selling opportunities. the ability to quickly harmonize data and unlock synergies has become a widely discussed challenge. Captures the typical data patterns and problems that arise after such mergers (using fully synthetic and anonymized datasets.)

Business Impact: Cost reduction,Efficiency improvements,Improved customer or user experience, Realizing cross-selling opportunities.